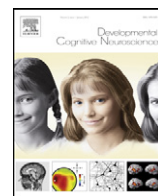




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Review

Neuronal effects following working memory training

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ABSTRACT

There is accumulating evidence that training working memory (WM) leads to beneficial effects in tasks that were not trained, but the mechanisms underlying this transfer remain elusive. Brain imaging can be a valuable method to gain insights into such mechanisms. Here, we discuss the impact of cognitive training on neural correlates with an emphasis on studies that implemented a WM intervention. We focus on changes in activation patterns, changes in resting state connectivity, changes in brain structure, and changes in the dopaminergic system. Our analysis of the existing literature reveals that there is currently no clear pattern of results that would single out a specific neural mechanism underlying training and transfer. We conclude that although brain imaging has provided us with information about the mechanisms of WM training, more research is needed to understand its neural impact.

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1. Introduction

In recent years, an increasing number of cognitive training studies have demonstrated not only improvements in the trained task but also in tasks that were not trained (see e.g. Lustig et al., 2009; Buschkuhl and Jaeggi, 2010; Klingberg, 2010; Shipstead et al., 2010; Morrison and Chein, 2011 for reviews). The common feature of many of these successful intervention studies is that training has focused on improving working memory (WM). WM is responsible for the active maintenance and manipulation of information for higher order cognition (Jonides et al., 2008). It has been shown that WM is predictive of a wide range of complex cognitive tasks such as planning or problem solving (Shah and Miyake, 1999), and also school-relevant tasks such as reading comprehension and mathematical skills (e.g. de Jonge and de Jong, 1996; Passolunghi and Siegel, 2001; Gathercole et al., 2006). In general, WM capacity is crucial for our general ability to acquire knowledge and learn new skills (Pickering, 2006), and it has been shown that WM is even better at predicting scholastic achievement than measures of intelligence (Alloway and Alloway, 2010). Deficits in WM are considered the primary source of cognitive impairment in numerous special-needs populations ranging from attention deficit hyperactivity disorder (ADHD) to mathematics disability (Minear and Shah, 2006). WM also has significant effects on classroom behavior. For example, children with lower WM capacity often forget teacher instructions, have difficulties staying on task in the classroom, and are easily distracted (Alloway et al., 2009). Teachers are more likely to rate children with poor WM capacity as disruptive and inattentive (Gathercole et al., 2006). In sum, WM is a crucial cognitive skill that is relevant for success in and out of schools. Given the relevance of WM to daily life and educational settings, it is not surprising that there is growing interest in the development of WM interventions, and current research suggests that WM training is indeed highly promising (see e.g. Lustig et al., 2009; Buschkuhl and Jaeggi, 2010; Klingberg, 2010; Morrison and Chein, 2011 for reviews). That is, it has been demonstrated that training on WM results in performance transfer to untrained tasks in younger and older adults (e.g. Buschkuhl et al., 2008; Jaeggi et al., 2008, 2010; Borella et al., 2010; Chein and Morrison, 2010), as well as children (e.g. Klingberg et al., 2005; Holmes et al., 2009; Thorell et al., 2009; Bergman Nutley et al., 2011; Jaeggi et al., 2011; Loosli et al., 2011). The reported effects can be very dramatic in that the transfer emerges in domains that are very different from the trained domain, such as reading (e.g. Chein and Morrison, 2010; Loosli et al., 2011), fluid reasoning (e.g. Klingberg et al., 2005; Jaeggi et al., 2008, 2010, 2011; Borella et al., 2010; Schmiedek, 2010), ADHD symptomatology (Beck et al., 2010), or drinking behavior (Houben et al., 2011). But there are other studies in which the transfer effects are observed only in tasks that are more closely related to the trained task, such as variants of the trained WM task (e.g. Holmes et al., 2009; Thorell et al., 2009; Bergman Nutley et al., 2011).

In a typical WM intervention study, participants train on a daily basis on one or several WM-based training tasks for 2–6 weeks. A common feature of almost all recent WM

training studies is the adaptive adjustment of difficulty of the training program. That is, the training task is made more difficult with increasing proficiency of the trainee, and it gets easier when the participant is not performing well on the task. In order to implement this adaptivity, almost all recent interventions are computer-based. The WM interventions that have been used vary in their complexity and range from repeating sequences of stimuli in the correct order to identifying targets in an *n*-back task in two modalities simultaneously.

The general procedure of almost all transfer-oriented cognitive training studies consists of at least two test sessions, one before and one after the intervention. In these sessions, performance on certain criterion tasks is measured in order to investigate whether the intervention has an impact on performance on these tasks.¹ That is, one is interested in whether the intervention was able to induce cognitive plasticity, defined as the acquisition of new cognitive skills (Willis and Schaie, 2009). When it comes to WM training, one is especially interested in process-based plasticity as opposed to plasticity that alters representations (Lustig et al., 2009; Lövdén et al., 2010). Additionally, participants are not introduced to any approaches or strategies that would help them perform the training task, but rather, participants repeatedly work on the training task on their own (Morrison and Chein, 2011). In order for the intervention to have an effect on the criterion measures, the training task has to be effective in that it must have the potential to induce cognitive plasticity. Furthermore, the training-induced plasticity must affect a general mechanism that at least partially underlies the training and the criterion task. This premise can easily be explained with an analogy from the physical domain: training of the cardiovascular system via running not only has a positive effect on running performance, but also on cycling performance (Suter et al., 1994). In this case, the cardiovascular system is the underlying mechanism that drives both running and cycling performance. While training effects with muscular skills are at least partially understood, the mechanisms underlying training effects for cognitive skills are still largely unknown. One promising way to explore such underlying mechanisms is by studying cortical changes or neural plasticity via neuroimaging (Poldrack, 2000; Willis and Schaie, 2009). It is the goal of this paper to review the current state of literature that investigates neural correlates of WM training with a special emphasis on studies that focus on transfer effects. However, in order to illustrate and discuss potential neural mechanisms that could result as a consequence of WM training; we also include intervention studies that are not based on WM training regimens. We do, however, limit our treatment to studies of humans in the interest of space limitations. We group the available studies into categories based on the methods to investigate neural mechanisms of training (see Table 1 for an

¹ There are studies that focus on so-called 'fatigue' effects or negative transfer, that is, lower performance on criterion tasks as a consequence of an intervention (e.g. Persson et al., 2007). We do not include such studies here. Instead, we focus on studies that aimed to find training-induced improvements on criterion tasks.

Table 1

Overview of studies that investigated neural effects as a consequence of WM training (healthy populations only).

Examined neural effect	Study	N	Age (years)	Control group	Transfer assessed	Method	Intervention	Training time	Main outcome
Functional activation changes	Garavan et al. (2000) – Exp. 1	12	26	No	No	BOLD	Visuospatial WM	20 min	Activation decrease
	Garavan et al. (2000) – Exp. 2	5	28	No	No	BOLD	Visuospatial WM	3 h 30 min ^a	Activation decrease
	Jansma et al. (2001)	15	23 (SD = 2.1)	No	No	BOLD	Sternberg item recognition task	45 min	Activation decrease
	Landau et al. (2004)	10	22–27	No	No	BOLD	Delayed face recognition task	30 min	Activation decrease
	Landau et al. (2007)	11	23.5	No	No	BOLD	Delayed object/spatial location recognition task	60 min ^a	Activation decrease
	Sayala et al. (2006)	10	20–34	No	No	BOLD	Delayed object/spatial location recognition task	<60 min	Activation decrease
	Schneiders et al. (2011)	48	23.7	Yes	No	BOLD	Visual and auditory <i>n</i> -back	10 sessions (8 h total)	Activation decrease
	Hempel et al. (2004)	9	26–32	No	No	BOLD	Spatial 0-back, 1-back, 2-back	4 weeks, 2×/day (total NA)	Activation increase then decrease
	Westerberg and Klingberg (2007)	14	24.9	Yes	Yes	BOLD	3 different WM tasks	5 weeks, 24.9 sessions (total NA)	Activation increase
	Jolles et al. (2010)	29	22.2 (19.3–25.3)	Yes	Yes	BOLD	Object span task (forwards and backwards)	6 weeks, 2.7×/week (6.9 h total)	Activation increase
	Dahlin et al. (2008) – Exp. 2	19	68.3 (65–71)	Yes	Yes	BOLD	6 different updating tasks	5 weeks, 3×/week (11.3 h total)	Activation increase
	Dahlin et al. (2008) – Exp. 1	22	23.6 (20–31)	Yes	Yes	BOLD	6 different updating tasks	5 weeks, 3×/week (11.3 h total)	Activation redistribution
	Olesen et al. (2004) – Exp. 2	8	29.3 (SD = 2.1)	No	Yes	BOLD	3 different visuo-spatial WM tasks	5 weeks, 18 sessions (total NA)	Activation redistribution
	Mozolic et al. (2010)	48	69.3	Yes	No	ASL	Interference paradigm	8 weeks, 1×/week (8 h total)	Inconclusive
Changes in cerebral blood flow at rest	Takeuchi et al. (2011b)	55	21.7	Yes	Yes	VBM	Mental calculation	5 days within 6 days (4 h per day)	Reduced gray matter volume
Structural changes	Takeuchi et al. (2010)	11	27.1 (SD = 1.4)	No	No	DTI	3 different WM tasks (visuo-spatial, <i>n</i> -back)	2 months, 25 min/session (total NA)	Increased white matter integrity
Dopaminergic function	McNab et al. (2009)	13	20–28	No	Yes	PET	10 different WM tasks (visuo-spatial, verbal)	5 weeks, 35 min/session (14 h total)	Change in D1 binding potential
	Bäckman et al. (2011)	20	19–33	Yes	Yes	PET	6 different updating tasks	5 weeks, 3×/week, 45 min/session	Increased Dopamine release during training performance
	Brehmer et al. (2009)	29	26 (20–31)	Yes	No	Genotyping	7 WM tasks (visuo-spatial, verbal)	4 weeks, 22.9 sessions (9.5 h total)	WM training performance modulated by genotype (trend only)
	Bellander et al. (2011)	29	26 (20–31)	Yes	No	Genotyping	7 WM tasks (visuo-spatial, verbal)	4 weeks, 22.9 sessions (9.5 h total)	WM training performance modulated by genotype

Notes: This table focuses only on studies that implemented a WM intervention.

^a Estimate, i.e., not reported by authors.

NA: not available.

overview). Finally, we conclude our review with a discussion of the implications for developmental and educational translation as well as with suggestions for potential future studies.

2. Neural mechanisms of transfer

We start with the assumption that for a training task to be effective, it must share at least some components with the outcome tasks (see Thorndike and Woodworth, 1901 for an early discussion of this topic). This might play out as a shared cognitive component. For example, a framework proposed by Halford et al. (2007) suggests that fluid intelligence (Gf) and WM share a common memory capacity constraint. So, one could conclude that training on WM might have a beneficial impact on Gf because of this shared cognitive component. An alternative way to think of overlap in components is in terms of shared neural mechanisms. The logic behind this idea is that training of a certain neural circuit might lead to transfer to other tasks that engage this circuit. Dahlin et al. (2008) conducted a study in which they trained young adults on six different updating tasks for five weeks. They found transfer to an untrained 3-back task but not to an untrained Stroop task. There were two critical differences between these two transfer tasks. First, the 3-back task required updating processes, whereas the Stroop task did not. Second, both the 3-back and the trained updating task activated the striatum, which was not the case for the Stroop task. The authors concluded that transfer can occur if the training and transfer task engage similar processes and brain regions. This result suggests that a neural association between the training and the transfer task is a necessary requisite for the training to result in transfer. To date, it is not clear how large the overlap of activated brain regions during training and criterion tasks has to be in order to result in transfer effects. Furthermore, it is not clear which processes and their neural correlates of training and criterion tasks must be similar in order to result in transfer. However, it makes intuitive sense that these overlapping processes and their associated neural correlates need to be trainable and they need to contribute substantially to training and criterion task performance in order to result in transfer.

In the following, we will discuss *how* brain regions can change as a consequence of training.

2.1. Changes in brain activation

There are four classes of activation patterns resulting from cognitive training that could emerge when examining functional changes (Jonides, 2004; Kelly et al., 2006). One is activation of the same brain regions before and after the intervention, but *decreased* activation of these regions after training. Such a pattern might reflect more efficient processing as a consequence of training, potentially leaving capacity for other processes. Another pattern is activation of the same brain regions before and after training, but with *increased* activation after training. This could be interpreted as an expansion of the neural structures involved in processing. The third pattern of results reflects a combination of the first two patterns, termed a redistribution of

activations. This reflects a combination of activation increases and decreases after training in the same areas that were activated before the intervention. Finally, the fourth pattern of results is activation in *different brain areas* after the intervention compared to baseline, reflecting a qualitative change in brain regions that are engaged by the trained task. Such a pattern might suggest that the intervention brought in new processes allowing new routes to task processing. In the following, we review the existing literature in the light of these four patterns of activation changes.²

2.1.1. Activation decreases

Activation decreases are primarily attributed to increased neural efficiency (Kelly et al., 2006). This notion is closely associated with the ‘neural efficiency theory’ postulated about 20 years ago by Haier et al. (1988, 1992a,b). The theory postulates that when participants are doing well on a task, they recruit fewer neurons than when they are not doing well; assuming that less proficient performance activates brain circuits that are inessential or even detrimental to task performance. For example, Haier et al. (1992a) found that 1–2 months of training with the computer game ‘Tetris’ led to decreased metabolic glucose levels in multiple brain areas. Although based on this early Positron Emission Tomography (PET) finding, the neural efficiency hypothesis is frequently discussed in relation to EEG work and is most often assessed with event-related desynchronization or synchronization of brain activity (ERD/ERS) in the alpha frequency band (between ~7 and 13.5 Hz; Klimesch et al., 2007). Within the framework of neural efficiency, it is usually assumed that higher alpha power (i.e., higher ERS) reflects increased neural efficiency (=less mental effort), which is related to cortical deactivation. Indeed, although somewhat inconsistent, there is cross-sectional work showing that higher intelligence is associated with higher alpha power (e.g. Neubauer et al., 1995; Jausovec, 1998), which is further strengthened by recent data investigating functional brain networks in relation to intelligence (Langer et al., 2011). In addition to cross-sectional work, there is at least one recent intervention study demonstrating increased ERS as a result of mental rotation training (Neubauer et al., 2010). A recent thorough review of the neural efficiency hypothesis is given by Neubauer and Fink (2009).

Functional MRI has also revealed decreased brain activation as a function of training (Garavan et al., 2000; Jansma et al., 2001; Landau et al., 2004, 2007; Sayala et al., 2006). The common feature of these WM training studies is that they used short practice periods lasting fewer than 4 h, administered in one session. Klingberg (2010) therefore concluded that reduced brain activations are most likely an exclusive feature of studies with such short practice times

² Most of the studies reported in this section used the blood-oxygen-level dependent (BOLD) effect as a method to quantify neural activation. Using the BOLD effect for imaging in training studies is not without controversy as, for example scanner drift can have unwanted influence on the functional data (e.g. Aguirre et al., 2002). However, a detailed analysis of imaging methods is beyond the scope of this paper, and we refer the reader elsewhere for methodological concerns (e.g. Aron et al., 2006).

and that the pattern in studies with longer training times might be more complex.

Nevertheless, a carefully conducted recent study using an adaptive *n*-back intervention found no activation increases after training, even though the intervention lasted over 8 1-h sessions (Schneiders et al., 2011). Although transfer effects were not assessed in this study, previous research using similar *n*-back training tasks has demonstrated beneficial effects in untrained tasks (Jaeggi et al., 2008, 2010, 2011). Following two weeks of either visual or auditory *n*-back training, Schneiders and colleagues observed activation decreases in the right superior middle frontal gyrus (BA 6) and posterior parietal regions (BA 40). Additionally, the more effective visual *n*-back training variant resulted in additional activation decreases in the right middle frontal gyrus (BA 9 and BA 46).

Further, in another *n*-back study conducted by Hempel et al. (2004), participants practiced on a 0-back, 1-back, and 2-back task for four weeks. Functional activation acquisition took place before training, after two weeks, and at the end of training. The authors reported an inverted U-shaped activation pattern in the intraparietal sulcus and the superior parietal lobe, indicating an activation increase during the first part of the practice period, followed by an activation decrease during the second part, which stands in direct contrast to the proposal by Klingberg (2010).

Another study by Dux et al. (2009) used various analytical tools to get at the underlying neural correlates of dual-task practice within a single study. In their intervention, participants trained on three different tasks in each session over a period of two weeks. The first task consisted of an auditory choice reaction paradigm, the second one of a visual choice reaction paradigm, and the last one consisted of a combination of the two tasks in which the auditory and the visual stimuli were presented simultaneously (dual task training). Although this study did not implement a WM intervention per se, and although it did not investigate transfer, we include it here because it adds to the picture of increased efficiency as a function of training in three important ways: First, the authors found no training-related activation increases in any brain region as a function of practice, but rather, a decrease of activation in the left inferior frontal junction. Second, by means of an effective connectivity analysis, they could show that this activation decrease was likely a result of local processes and not due to changes in interregional connectivity. And finally, by analyzing the time course of the blood-oxygen-level dependent (BOLD) signal before and after training, they found evidence that the decrease in activation most likely emerged due to an increase in speed of information processing in the brain region that was sensitive to training effects. Taken together, these three analyses suggest a more efficient neural network as a function of training; that is, the brain adjusted from slow and effortful, to fast and more automatic processing (Jansma et al., 2001; Chein and Schneider, 2005).

These findings are further supported by two longitudinal studies of second language learning (Stein et al., 2006, 2009). In these studies, increased second language expertise that was developed over the course of several months corresponded to reduced frontal brain activation

at post-test, and further, to increased processing speed and shorter brain activations in similar regions in which the activation decreases were found. Taken together, there seems to be accumulating and consistent evidence that neural processes can become more efficient as a function of practice and expertise, which is commonly reflected in activation decreases as observed with fMRI, or increased ERS as observed with EEG.

2.1.2. Activation increases

Increased brain activation is commonly observed as a consequence of practice on sensory or motor tasks (e.g. Elbert et al., 1995; Ungerleider et al., 2002). It reflects either a more extensive recruitment of cortical areas due to an increase in the size of cortical representations (Pascual-Leone et al., 2005), or a stronger neural response in existing areas (Kelly et al., 2006). Based on the skill acquisition literature, Westerberg and Klingberg (2007) hypothesized that practice on a WM intervention program that previously resulted in transfer to untrained tasks (Klingberg et al., 2002, 2005) might also result in practice-induced brain activation increases. Westerberg and Klingberg found such brain activation increases in the middle or inferior frontal gyrus as well as in the parietal cortex. However, generalization of these results is very limited since the effects are based on data of three participants. Furthermore, the results are only partially in line with the results from a prior study by this group (Olesen et al., 2004; see also below).

There is additional evidence for increases in brain activation following WM training from a recent study in which young adults practiced on a simple forward and backward object span task about three times a week over the course of 6 weeks (Jolles et al., 2010).³ Before and after the practice period, participants were scanned on the practiced task and the results were compared to those of participants who were scanned at the same time but who did not engage in the practice program. The results yielded differential outcomes for the forward (i.e., maintenance) and backward (i.e., manipulation) version of the task. Whereas task-related activation increases in default-mode regions were observed in the forward version, the backward version resulted in activation increases in the striatum and in the left ventrolateral prefrontal cortex.

Finally, in a study in which old adults were trained on different updating tasks for 5 weeks, only brain activation increases as a function of training were found (Dahlin et al., 2008; Experiment 2). However, this stands in contrast to the activation patterns that the authors observed in young adults with the same training regimen (Dahlin et al., 2008; Experiment 1; see below); thus, the activation increases observed in the old adults could reflect differential plasticity processes as a function of age (Lövdén et al., 2010).

In sum, in contrast to what has been found in the motor learning literature, it seems that the evidence for pure increases in brain activations after WM training is very scarce.

³ It should be noted in passing though that the authors did not find any transfer after their intervention.

2.1.3. Activation redistribution

Another pattern of results that might result from training is a combination of activation decreases and increase across a number of brain areas that are present before and after the intervention (Kelly et al., 2006). The idea behind this outcome is that on the one hand brain activation decreases might be found in regions that are typically responsible for more general processes such as attentional control (e.g. in the prefrontal cortex). On the other hand, there might be brain activation increases in areas that typically support task-specific functions. It is assumed that the brain areas supporting attentional control are especially used when coping with a novel task in the beginning, and that with increased training, the task becomes more automated and requires less attentional control, i.e., the required processes are more accessible and more efficient (Petersen et al., 1998; Jansma et al., 2001; Chein and Schneider, 2005).

It is exactly this pattern of results that is reported by Dahlin et al. (2008; Experiment 1). In this study, young adults trained on different updating tasks over a period of 5 weeks. The authors reported increased activity mainly in the striatum but also in temporal and occipital brain regions, as well as decreased brain activity in frontal and parietal regions. In accordance with the framework put forward by Petersen et al. (1998), the frontal activation decreases might reflect more automatic processes as a result of practice. However, it is harder to explain the parietal brain activation decreases with this framework.

In another study, young adults were trained on three visuo-spatial WM tasks for five weeks (Olesen et al., 2004; Experiment 2). A comparison of brain activations before and after the intervention revealed increased brain activity in frontal and parietal brain regions, as well as in the basal ganglia and the thalamus; regions that are commonly involved in WM processes (Jonides, 2004). But Olesen and colleagues also reported decreased brain activation in brain regions such as the anterior cingulate, the postcentral gyrus, and the inferior frontal sulcus. Especially the anterior cingulate is often associated with attentional control and mental effort (Bush et al., 2000). Thus, it might be that the task requires less attentional control after training, resulting in a decrease in activation as proposed in Petersen's framework (1998). But again, the activation decreases in the other regions are not easily explained with this framework.

To conclude, it seems that WM training can result in a quantitative shift in brain activation. However, future work needs to confirm the tentative evidence provided by the two studies reported above.

2.1.4. Reorganization of networks

This final category of potential outcomes concerning brain activation changes refers to a reorganization of processing, leading to a qualitative change in the processes used to accomplish the trained task (Kelly et al., 2006). In contrast to activation decreases and increases for which it is assumed that the same processes are still at work after an intervention at a quantitatively different level, it is assumed that a reorganization reflects a qualitative cognitive and neural change as a consequence of training. For example,

Erickson and others trained their participants on a single and on a dual-task paradigm over a period of 2–3 weeks (Erickson et al., 2007). Comparison of brain activations assessed before and after the intervention revealed mostly reductions in brain activations such as in the ventrolateral prefrontal cortex. However, in the dual-task condition, a whole-brain voxelwise analysis revealed a significant time (pre vs. post) $\times$ group (experimental vs. control) interaction in a region located in the dorsolateral prefrontal cortex (DLPFC) bilaterally. Inspection of the parameter estimates of the peak voxels in these areas revealed that the activation was statistically not different from zero in the pre-test but reached threshold in the post-test. Further Analyses of Variance (ANOVA) confirmed that the observed pre to post-test difference indeed reflected an increase in activation and not a reduction in deactivation. In sum, although there is quite a body of literature available that has reported reorganized networks following various regimens of cognitive training (Raichle et al., 1994; Poldrack et al., 1998; Fletcher et al., 1999; Petersson et al., 1999; Poldrack and Gabrieli, 2001), to our knowledge, there is no such report of a WM training study that demonstrated this pattern of activation change.

2.2. Resting state changes

The brain activation changes we have discussed so far all address task-induced responses. However, it is also of interest to consider the intrinsic activity of the brain that emerges due to environmental interactions (Raichle, 2010). One way to examine such intrinsic activation is to study the functional connectivity of brain regions when the participant is at rest and does not engage in goal-directed activity (Peltier and Noll, 2002; Greicius et al., 2003). Another way is to measure cerebral blood flow (CBF) at rest with a quantitative imaging method such as PET (Phelps and Mazziotta, 1985), or Arterial Spin Labeling (ASL) (Detre et al., 2009). The question that arises, regardless of method, is whether training on WM has the potential to result in resting state brain changes even if the participant is not engaging in task-related activities, whether during training or transfer. Demonstrating such effects would provide very compelling evidence for a mechanism that underlies generalized improvement because it could reflect profound changes in neural processing as a function of training.

2.2.1. Functional connectivity

When not engaged in goal-directed behavior, the brain demonstrates a certain pattern of activations that is termed default-mode network (DMN) activity (Raichle et al., 2001). It has been shown that there is a strong correlation among the nodes within this network, supporting the notion that there is a functional coherence in the DMN (Greicius et al., 2003). There is cross-sectional research demonstrating that the functional connectivity (that is, the correlation within activated areas of the DMN) correlates with WM capacity (Hampson et al., 2006), as well as intelligence (Song et al., 2008). Therefore, in that individual differences in WM capacity and intelligence seem to be related to differences in resting state (i.e., task-independent) connectivity, it is conceivable to hypothesize that individual differences

in task proficiency (as obtained by training) might also lead to changes in resting state. To our knowledge, there are no WM intervention studies available that investigated functional connectivity changes. However, there are two studies that demonstrate that cognitive intervention and experience can have an impact on functional connectivity (Lewis et al., 2009; Jang et al., 2011).

Lewis et al. (2009) let participants practice on a perceptual learning task which required identification of a target stimulus among distractor items. Interestingly, they observed significant functional connectivity differences when comparing connectivity measurements taken before and after the intervention. They found negative correlations between trained visual brain regions and frontal as well as parietal areas after practice. Notably, before practice, these regions did not correlate with each other. Lewis and colleagues assumed that these negative correlations reflect more efficient processing because of reduced interference between these brain areas, thus again, providing evidence for the neural efficiency hypothesis described earlier. Additionally, the authors reported a negative correlation between the visual cortex and the brain regions related to the DMN. This finding is remarkable in that it demonstrates a relationship of the DMN to perceptual functions, which adds to the widely assumed self-referential nature of the DMN (Christoff et al., 2009).

Another recent study published by Jang et al. (2011) compared DMN activity of long-term meditation practitioners with a group of healthy controls. Their results revealed increased functional connectivity in the anterior medial prefrontal cortex (MPFC) in the meditation group. Since the MPFC is associated with self-relevant mental simulations, the authors argue that repetitive meditation practice may have led to increased neuronal connectivity in this area, which in turn led to the increase in functional connectivity.

Based on the studies reported above, we believe that it is not unrealistic to expect that WM training also alters functional connectivity.

2.2.2. Cerebral blood flow at rest

Another potential mechanism underlying cognitive interventions is of a vascular nature. Comparable to practice effects in skeletal muscles which usually result in an increased blood flow at rest (Flück and Hoppeler, 2003), it might be possible that cognitive training results in a 'fitter' brain, a hypothesis which could be assessed by measuring CBF at rest. In a PET study that involved a brief 30 min reasoning intervention between two scanning sessions, increased CBF was found in occipital, superior temporal, and ventromedial prefrontal cortices, as well as in the pulvinar at post-test (Mazoyer et al., 2009). The authors argue that the increase in CBF might be due to an increase in synaptic activity, which in turn augmented the metabolic demand in these areas. However, the shortness of the intervention suggests that the reported effects might be rather transient in nature and most likely do not reflect a lasting alteration of cerebral perfusion. Mozolic et al. (2010) trained old adults once a week for about 1 h over a time period of 8 weeks on an interference and attentional control paradigm. ASL was

used prior to and after the intervention in order to quantify cerebral perfusion. The authors reported an increase in CBF in the right inferior frontal cortex (IFC) in the trained group. Unfortunately, however, there was a significant difference in CBF at baseline in the IFC between the trained and the control group in that the training group had much lower CBF at pre-test, and therefore, the reported increase in cerebral perfusion in this region might reflect a regression to the mean effect.

In sum, the findings concerning cerebral perfusion changes following cognitive interventions are inconclusive to date.

2.3. Gray and white matter changes

One might imagine that there are changes in brain structures that precede functional changes, or alternatively, that there are changes in brain morphology that result from functional alterations. Both alternatives would provide further insight into brain plasticity, because changes in brain morphology could be seen as the strongest evidence for lasting intervention effects. There is ample evidence from post-mortem and in vivo studies demonstrating changes in the structural architecture of the human brain over the lifespan, with the general pattern showing an inverted U-shaped pattern of development in gray matter volume, and a linear increase in white matter volume and density up until young adulthood, after which there is a decrease with normal aging; developmental changes that seem to reflect experience and learning processes to a large extent (see e.g. Casey et al., 2005 for a review). However, relatively little is known as to whether and how interventions that only last for several weeks can significantly alter the human brain structure in similar ways as lifelong maturation and development does.

To date, the most frequently applied method to track and characterize structural changes in vivo is voxel-based morphometry (VBM), which makes use of structural MR images to quantify gray matter, white matter, and cerebrospinal fluid (Mechelli et al., 2005). VBM allows the investigation of the living brain in contrast to earlier studies that had to rely on post-mortem analyses, thereby making VBM a valuable tool to investigate how brain morphology changes as a function of learning.

There is a vast number of cross-sectional and correlational studies showing that brain morphology is associated with learning and experience in different domains. It has been shown that aspects of brain structure are correlated with various levels of expertise in the domains of navigation (Maguire et al., 2000), music (Sluming et al., 2002; Gaser and Schlaug, 2003; Bengtsson et al., 2005; Schlaug et al., 2005; Han et al., 2009), meditation (Lazar et al., 2005; Pagnoni and Cekic, 2007; Hölzel et al., 2008; Luders et al., 2009; Vestergaard-Poulsen et al., 2009), or mathematical skills (Aydin et al., 2007). The general consensus is that expertise is associated with larger gray matter density (or volume) and/or with stronger white matter integrity in those brain regions that are involved in the respective criterion task. Unfortunately, cross-sectional studies have the limitation that they are not able to prove causality and that learning effects cannot be disentangled from other

confounding factors such as environmental and/or genetic influences.

Nonetheless, there are a few experimental studies which provide evidence for changes in brain morphology as a result of relatively short-term cognitive interventions (see e.g. Draganski and May, 2008 for a more general review on the effects of experience, including motor skill learning on brain structure). Although most of these studies did not rely on WM based interventions, they are nevertheless relevant in that they examine changes in brain structure as a function of an intervention targeting higher cognitive functions.

There are two published studies available that investigated gray matter volume (GMV) of medical students before and after they had been studying for their medical examinations. The study groups' GMV was compared to a group that did not study (Draganski et al., 2006; Ceccarelli et al., 2009). Both articles reported increases in GMV in several brain regions after the study phase, including the left dorsomedial frontal cortex, right orbitofrontal cortex, and left precuneus (Ceccarelli et al., 2009), posterior and inferior parietal cortices bilaterally, as well as the posterior hippocampus (Draganski et al., 2006). In addition, Draganski and colleagues also found a decrease in GMV in the occipital lobe, however, the effects in this region were close to a region where the authors observed a significant increase in white matter volume, which seems to have prompted this inverse effect (Golestani et al., 2002; Draganski et al., 2006). The differential results in terms of localization between the two studies might lie in the difference in study phases (two weeks vs. three months), and potentially, in the different methods used to assess GMV and white matter volume (tensor-based morphometry (Ceccarelli et al.) vs. VBM (Draganski et al.)).

Another recent study investigated the effects of an intensive reading intervention on GMV in a group of children with dyslexia (Krafnick et al., 2011). Consistent with the studies summarized before, the authors observed increases in GMV after an 8-week training period in the left anterior fusiform gyrus, right hippocampus, left precuneus, and right cerebellum, which were accompanied by improvements in reading performance. There were no further increases (or decreases) in GMV over an 8-week follow-up period during which there were no further gains in reading performance. Although there was no control group, the data still suggest that the increases in GMV emerged as a result of training because they were measurable only when participants were engaged in the intervention, not eight weeks after training completion.

Ilge et al. (2008) investigated the effects of a 2-week practice period of mirror reading. This study is of interest here in that it combines functional and structural brain data in order to investigate whether there are overlaps between the regions which show changes in activation and those which show changes in GMV. Indeed, the authors found an activation decrease as a function of training in the right superior parietal cortex, but also an activation increase in the right dorsal occipital cortex. In addition, there was an increase in GMV in the right occipital cortex, and importantly, this increase in GMV overlapped with the peak of the activation increase as a function of mirror reading practice,

suggesting at least some coherence between the changes in task-specific activation and changes in structure.

Finally, a study by Takeuchi et al. (2011b) is the only one to our knowledge that investigated GMV changes following WM training. In their study participants trained on a task that consisted of calculating mental addition and multiplication problems over a period of 5 days for 4 h a day. Somewhat surprisingly, the authors reported reductions exclusively in GMV that were located in the bilateral DLPFC, right inferior parietal lobule, left paracentral lobule, and left superior temporal gyrus. The finding is somewhat unexpected as there is currently no theory to explain why learning in adults would result in reductions in GMV (Draganski et al., 2006).⁴ However, the reported results by Takeuchi et al. (2011b) are consistent with a very recent study by the same group in which they also found reductions in GMV but this time as a consequence of training processing speed (Takeuchi et al., 2011a). Nevertheless, it might have been that increases in white matter density could account for the regional loss in GMV (Golestani et al., 2002; Draganski et al., 2006); a potential underlying mechanism that was not tested in this study.

In another study, Takeuchi et al. (2010) investigated changes in white matter microstructure by means of another MR-based technique, i.e., diffusion tensor imaging (DTI). DTI provides measures of magnitude and direction of water diffusion in the brain tissue. Fractional anisotropy (FA) is a commonly used diffusion parameter, which quantifies the directionality of the diffusion (Mori and Zhang, 2006). It is assumed that the magnitude of FA reflects the anatomical features of white matter, such as axonal membrane thickness and diameter, fiber density, and myelination (Scholz et al., 2009). Therefore, FA is interpreted as a measure of the quality of white matter fiber tracts, or axonal integrity and coherence (Alexander et al., 2007). As a result of a 2-month WM training period, Takeuchi and colleagues observed an increase in FA in parietal as well as frontal cortices. Unfortunately, no control group was included in the design, making it difficult to determine the causality of the FA increases. However, the number of completed training sessions correlated positively with pre- to post-training increases in FA, suggesting at least some neural effects as a consequence of the intervention. Further, a control group used by Scholz et al. (2009) that was scanned in a comparable interval but that did not engage in any intervention did not show any changes in FA. Finally, consistent with the findings referenced above, Tang et al. (2010) observed increases in FA as a function of a short-term (11 h) meditation intervention in tracts that connect regions that are involved in attentional control such as the anterior cingulate cortex (ACC) and parietal cortices; networks which are correlated with behavioral effects of meditation practice (Tang et al., 2007). It is important to note that the increase in FA in the study of Tang et al. was only observed in the experimental group, but not the active control group, thus, it seems likely that the

⁴ In the developmental literature, gray matter loss is usually interpreted as a result of synaptic pruning and/or an elimination of connections reflecting fine-tuning of cortical networks (cf. Casey et al., 2005).

FA increases observed in Takeuchi and colleagues' study reflect specific effects of the WM intervention.

Despite the converging evidence that there are observable changes in GMV and white matter integrity as a function of relatively short-term interventions, the exact cellular mechanisms underlying these changes as well as their potential impact on the MR signal are only poorly understood (Draganski and May, 2008). It has been speculated that changes in GMV might be indicative of dendritic spine growth leading to the sprouting of new connections and/or a modification of existing connections. It is assumed that such intracortical axonal changes in neural architecture can happen relatively fast in a time frame of days to weeks (Trachtenberg et al., 2002). And indeed, results of a study by May et al. (2007) revealed that GMV changes are visible after as little as 5 days of repetitive transcranial magnetic stimulation (May et al., 2007). A similar time frame for morphological changes was later confirmed in a behavioral study in which 7 days of juggling training resulted in GMV increases (Driemeyer et al., 2008). Although these results suggest dendritic spine growth and associated synaptogenesis as neurobiological mechanisms of GMV changes, they do not rule out other, presumably slower, mechanisms such as neurogenesis, genesis of glial cells, increase of cell sizes, or angiogenesis (e.g. Swain et al., 2003; Bailey et al., 2004; Kempermann et al., 2004).

Finally, the changes in FA might reflect changes in myelination, resulting in an increased conduction velocity and signal synchronization which might facilitate the communication among neural circuits, which can in turn determine observed performance improvements (Scholz et al., 2009; Takeuchi et al., 2010; Tang et al., 2010). Other possible mechanisms could be a change in density and/or in the diameter of axons (Scholz et al., 2009). Finally, as glial cells seem to play an important role in inducing structural changes in both gray and white matter, there is the possibility for a common underlying mechanism driving structural changes and, as such, learning-related plasticity (Scholz et al., 2009).

As in most of the intervention studies we have reviewed, the exact time course of the structural changes is not known. We do not know how long an intervention has to last to yield structural changes, we do not know how long such changes last, and finally, we do not know whether there are differential underlying mechanisms as a function of time. There are studies that suggest that at least three months are needed for stem cells to differentiate into neurons, which was the reason that Draganski and May (2008) had their participants train for 3 months. However, there are intervention studies that demonstrate changes in GMV in as little as one week (May et al., 2007; Driemeyer et al., 2008). And finally, there have been reports suggesting that myelination can be changed by electrical activity in a mere couple of days (Demerens et al., 1996; Ishibashi et al., 2006). Another open question is the relationship and the directionality of the structural changes and the changes in functional reorganization, whether function precedes structure or whether structure determines function. Although one study that combined VBM and fMRI methods has done some work in this direction (Ilg et al., 2008), more studies are clearly needed to clarify this issue.

2.4. Dopamine-related changes

Another road to get at physiological correlates of WM training was taken by McNab et al. (2009) who examined dopamine D1 and D2 receptor density before and after training. It is widely known that dopamine (DA) plays an important role in WM and is related to higher cognitive functions (see e.g. Goldman-Rakic, 1998; Nieoullon, 2002 for reviews), and thus, it seems likely that there is an association between the functioning of the dopaminergic system and the effects of WM training. Indeed, McNab et al. report a negative correlation between training-related performance increase and changes in cortical D1 binding potential in prefrontal and parietal cortices, which mainly resulted from a decrease in D1 receptor density. This finding is consistent with earlier studies demonstrating a negative correlation between D1 binding and WM capacity in Schizophrenia (Abi-Dargham et al., 2002).

In a very recent study, Bäckman et al. (2011) investigated DA activity as a consequence of the same updating training regimen this group used before to investigate activation changes (Dahlin et al., 2008). Similar to the results reported by McNab et al., they found reduced DA binding potential, however, not in D1, but D2 receptors. The changes were found exclusively in the left caudate, the same brain regions where activation changes were found before, using the same intervention (Dahlin et al., 2008).

The underlying mechanisms and the causal role of such decreases in receptor density are currently not clear. However, there are two recent studies providing converging evidence that DA plays a causal role in training-related performance increases in that they investigated the role of certain genotypes on WM training (Brehmer et al., 2009; Bellander et al., 2011). Although both studies presumably relied on the same participants and suffer from a small sample size ($N=29$), both indicate that certain genetic variations might be associated with training gains in WM over time, suggesting an advantage in the ability to learn. One of the studies investigated polymorphisms of the DA transporter (DAT1) gene, a gene that seems to be essential for regulating DA availability in the striatum (e.g. Goldberg and Weinberger, 2004). Although only a tendency, the data suggest an advantage of the 9/10-repeat carrier group compared to the 10-repeat carriers in terms of training gain on WM. The other study focused on differential effects of variations in the LIM homeobox transcription factor 1 alpha (LMX1A) on WM training performance. It has been shown that LMX1A plays an important role in DA development and regulation in the midbrain (e.g. Prakash and Wurst, 2006), which again warrants the hypothesis that there might be differential effects on WM as a function of LMX1A genotype. As in the previous study, there were no group differences at baseline in any of the tasks, however, there were significant training effects as a function of genotype in one of the LMX1A single nucleotide polymorphisms (SNPs) that the researchers assessed (i.e., in rs4657412): The TT carrier group outperformed the CC/CT carrier group in one of the training tasks overall, and in addition, although the TT carrier group started at a higher level, the training gain was still larger in this group; thus, there was no evidence for a regression to the mean effect.

In sum, there seems to be some preliminary evidence from a few studies suggesting a causal role of DA on WM training performance.

3. Conclusion and future directions

There is no doubt that structural and functional brain changes take place in the course of typical human development (e.g. Casey et al., 2005). It has also been argued that such changes go beyond the developmental maturation period and that plasticity is an inherent property of the human brain (Pascual-Leone et al., 2005; Draganski and May, 2008; Lövdén et al., 2010). Our review certainly adds to this notion and sheds some light on the nature of these plasticity processes. However, despite the promising results in the behavioral training literature, the research regarding the underlying neural correlates has been slowly emerging in the last couple of years. With the few studies published on this topic, there is currently no clear pattern of results that would single out a specific neural mechanism underlying training and transfer that would fit within one single framework – there is evidence for activation increases, activation decreases, and a combination of both. Further, there are changes in resting-state connectivity, as well as structural changes and changes in dopaminergic function that may or may not directly relate to the functional changes. Finally, there is no single brain region that seems to be consistently involved in every training intervention; rather, activation and structural changes seem to be largely task-dependent.

It is very likely that part of the inconsistent pattern of findings is due to the different methods and tasks applied, but also to certain factors that might moderate some of the effects. Several moderators have already been identified behaviorally and most likely also affect the neural effects, such as training length (Hempel et al., 2004; Basak et al., 2008; Jaeggi et al., 2008), task difficulty or task demands (Jolles et al., 2010), individual differences in pre-existing ability and/or individual differences in training performance (Bellander et al., 2011; Jaeggi et al., 2011), gender (Neubauer et al., 2010), age (Dahlin et al., 2008; Karbach and Kray, 2009; Schmiedek, 2010), or even motivational processes such as the participants' effort directed to the task (Neubauer and Fink, 2009; Jaeggi et al., 2011). Future research should therefore carefully consider possible moderators, which of course requires that researchers conduct their studies with adequate sample sizes in order to allow the detection of such effects. Further, knowledge about those moderators should ultimately allow researchers to design specific interventions that directly target particular cognitive skills on an individual level with the ultimate goal to increase the efficacy of the intervention.

Also, it would be beneficial for future research to apply multiple methods of investigation, that is, to assess behavioral performance and combine those data with imaging data such as fMRI and VBM and/or DTI, as well as EEG, or to apply multiple analytical tools in order to be able to distinguish among several possible neural mechanisms that could account for training and transfer effects. To date, there are very few studies that have attempted to do this (Ilg et al., 2008; Dux et al., 2009).

Of course, in the long run, it is important to get at the neural correlates not only of the training task, but also of the transfer task. The logic behind this is that one should examine networks of activation that are involved in both training and transfer tasks, and to investigate how these networks change as a function of the intervention. Unfortunately, to our knowledge there is currently only one study that has attempted to do this (Dahlin et al., 2008), and clearly more data are needed to forward our understanding of the underlying mechanisms of training, and ultimately, transfer. As we mentioned earlier, there is a growing body of literature that investigates transfer effects in children following cognitive training, especially WM training (e.g. Klingberg et al., 2002, 2005; Holmes et al., 2009; Thorell et al., 2009; Bergman Nutley et al., 2011; Jaeggi et al., 2011; Loosli et al., 2011), and it has been concluded that such interventions have clear effects on executive control, processes that are crucial for school readiness and academic success (see Diamond and Lee, 2011 for a recent review). However, to date, there are no studies available that have examined neural correlates of transfer effects in children. Nevertheless, the adult training studies do allow one to speculate about implications for developmental and educational research.

For example, it has been repeatedly shown that there are differential patterns of brain activations between children and young adults. For example, one study demonstrated that 8 to 12 year old children did not activate the DLPFC and the superior parietal cortex during a delay period of a WM task in contrast to young adults who did activate these brain areas (Crone et al., 2006). Furthermore, brain activations in the right DLPFC positively correlated with task performance in the study reported by Crone and others. More generally, a review article focusing on neural development and WM performance concluded that activation increases in task-related brain areas – especially in frontal, parietal, and striatal regions – are associated with improved WM performance (Bunge and Wright, 2007; see also Rypma and Prabhakaran, 2009). As suggested by Crone et al. (2006), it might be that children start to recruit task-relevant brain areas earlier when they are trained on tasks that rely on those brain areas. Such training would essentially allow typically developing children to get a head start in their WM abilities. Probably more importantly, one could also imagine that children with developmental deficits can benefit from such training in that it would allow them to bring potential immature neural activations on par with activations as measured in typically developing children. Indeed, it has been shown that WM interventions are most promising in children with deficits in that domain, such as ADHD. Thus, WM training seems to be a relatively easy and accessible means to reduce the achievement gap (Diamond and Lee, 2011), which ultimately, should also be observable on a neuronal basis.

A potential candidate for such a training task could be an *n*-back task that has resulted in generalized behavioral benefits in typically developing children (Jaeggi et al., 2011) and which has been shown to consistently activate DLPFC as well as parietal regions in adults (Owen et al., 2005). To date, there are three studies that investigated neural consequences as a result of *n*-back training (Hempel et al., 2004;

Takeuchi et al., 2010; Schneiders et al., 2011). Based on the results as reported by Hempel et al. and Schneiders et al., training on *n*-back results in activation decreases, especially in the long run, and affects DLPFC as well as parietal regions. Therefore, it might well be that the generalized improvements, as reported by Jaeggi et al. (2011), are a result of activation changes in prefrontal and parietal brain regions.

In sum, despite the few studies that investigated the neural correlates of WM training, the results suggest a dynamic pattern of functional and structural plasticity underlying experience and learning. The future challenge will be to determine the exact mechanisms so they can be translated into treatment and educational practice.

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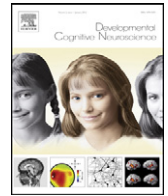
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Update

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Erratum

Erratum to: Supplement: Neuroscience & Education

In the 2012 *Developmental Cognitive Neuroscience* supplement 'Neuroscience & Education' (Volume 2, Supplement 1, ppS1–S204) the conflict of interest statements were missing from the following papers. The authors' conflict of interest statements are listed below.

Early adversity and neural correlates of executive function: Implications for academic adjustment

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Brain function during probabilistic learning in relation to IQ and level of education

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Neural circuitry associated with two different approaches to novel word learning

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Early math achievement and functional connectivity in the fronto-parietal network

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Weak task-related modulation and stimulus representations during arithmetic problem solving in children with developmental dyscalculia

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Neuronal effects following working memory training

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Practice effects in the developing brain: A pilot study

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Enhanced efficiency of the executive attention network after training in preschool children: Immediate changes and effects after two months

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